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AI for Bharat Hackathon

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Team Name : Spirit

Team Leader Name : Meghna Subramani

Problem Statement : Build an AI-powered solution that improves access to information, resources, or opportunities for communities and public systems.

Brief about the Idea:

- **The AI Emergency Help Assistant is a multilingual, voice-enabled, low-bandwidth AI system designed to support people during the first critical minutes of an emergency—before ambulances, police, or fire services arrive.**
- **It detects emergencies automatically, identifies the type of emergency, and provides step-by-step first-aid guidance, while also helping users contact nearby emergency services and SOS contacts.**
- **The solution is optimized for panic situations, rural or low-connectivity areas, and users with little medical knowledge.**

Problem Definition

In real emergencies (collapse, accident, disaster):

- **People panic and forget basic first aid**
- **Emergency numbers are not easily available**
- **Users don't know what details to tell responders**

Existing apps are text-heavy, English-centric, and require stable internet

- **How different is it from any of the other existing ideas?**

Existing Solutions	AI Emergency Help Assistant
Static emergency apps	Dynamic AI-driven guidance
General health chatbots	Emergency-only intelligence
English-first	Local language + Voice
Internet dependent	Low-bandwidth / offline fallback
No panic handling	Step-by-step calm coaching

- **How will it be able to solve the problem?**

1. **Detects emergency intent from voice/text**
2. **Classifies emergency type (medical, accident, disaster, threat)**
3. **Immediately switches to emergency mode (no long explanations)**
4. **Guides step-by-step actions (first aid, safety steps)**
5. **Fetches nearest emergency services**
6. **Prepares responder-ready information**
7. **Works offline if needed**

This reduces panic, saves time, and improves survival probability.

- **USP(Unique Selling Proposition) of the proposed solution**

"Life-saving guidance in the first 5 minutes, for anyone, anywhere."

1. **Emergency-first AI (not a general chatbot)**
2. **Voice-guided first aid coaching**
3. **Local language + cultural context**
4. **Works in low connectivity environments**
5. **Tells users *what to say* to emergency responder**

Why AI is **Required** in the Solution

AI plays an important role in quickly understanding emergency situations and providing the correct guidance AI helps the system to:

- Detect emergency intent from voice or text
- Classify the type of emergency (medical, accident, disaster, threat)
- Provide real-time step-by-step instructions
- Communicate in multiple languages
- Support users in panic situations with calm guidance
- Without AI, the system would only provide static information and would not be able to respond dynamically to emergencies.

How AWS Services are Used in the Architecture

AWS cloud services help the system run reliably and scale for many users.

Example AWS services used:

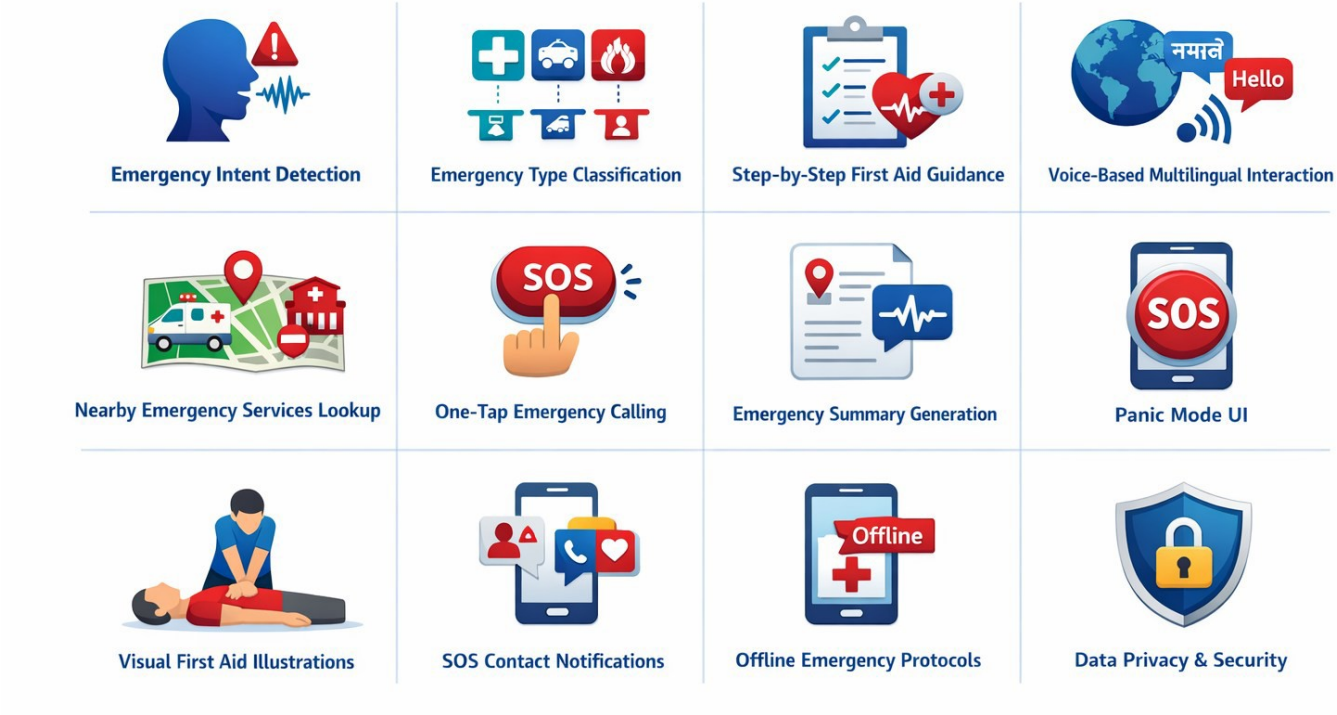
Amazon Lex – for voice interaction and chatbot interface
Amazon Transcribe – converts speech to text
Amazon Polly – converts AI responses into speech
AWS Lambda – processes emergency requests and logic
Amazon DynamoDB – stores emergency protocols and user session data
Amazon S3 – stores training data and system resources
Amazon Bedrock / SageMaker – AI models for intent detection and classification
Amazon Location Service – finds nearby hospitals and emergency services
These services ensure the solution works efficiently even during high demand.

What value the AI layer adds to the user experience?

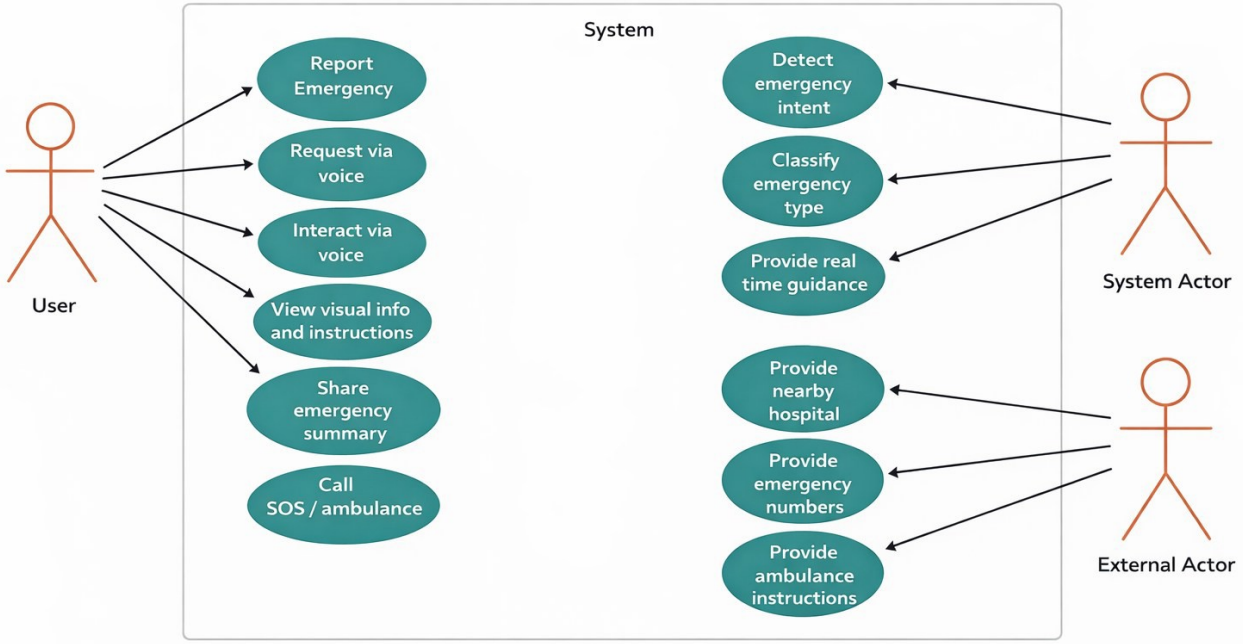
Value the AI Layer Adds to the User Experience.

- Instant emergency detection.
- Real-time voice guidance.
- Real-time voice guidance Multilingual communication.
- Personalized instructions based on the situation.
- Faster response during the critical “golden minutes”.
- This improves user confidence and increases the chances of saving lives.

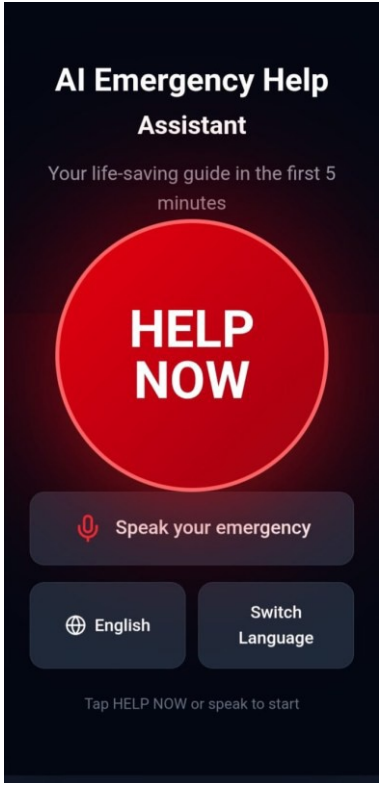
List of features offered by the solution

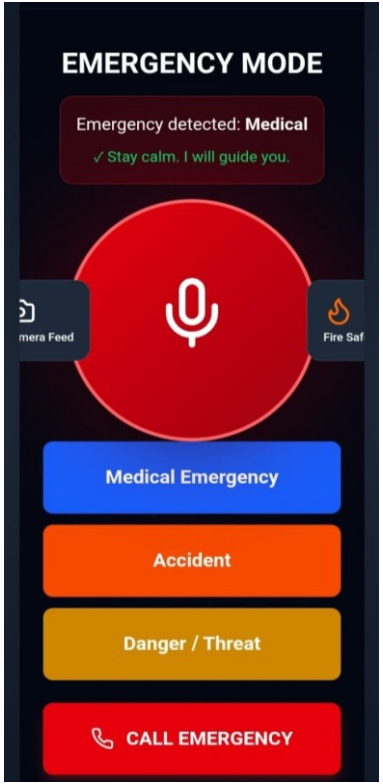


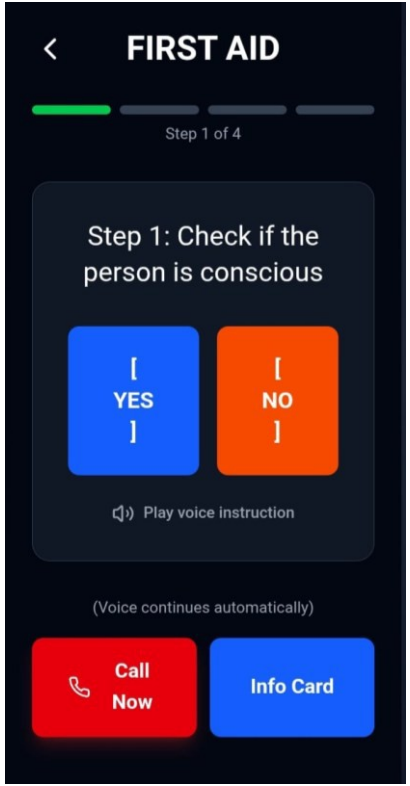
Process flow diagram or Use-case diagram – AI Emergency Assistant

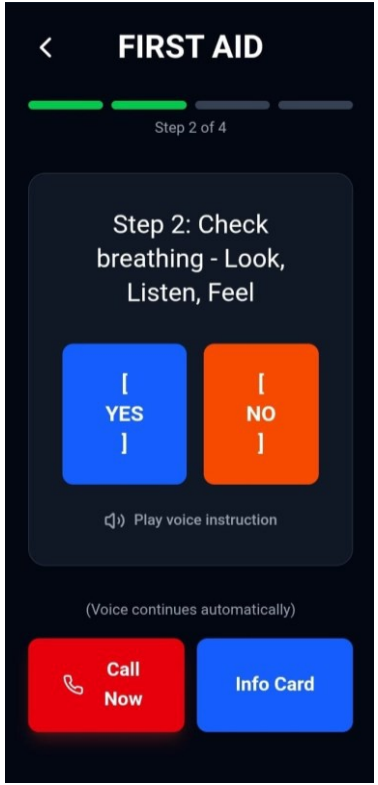


Wireframes/Mock diagrams of the proposed solution









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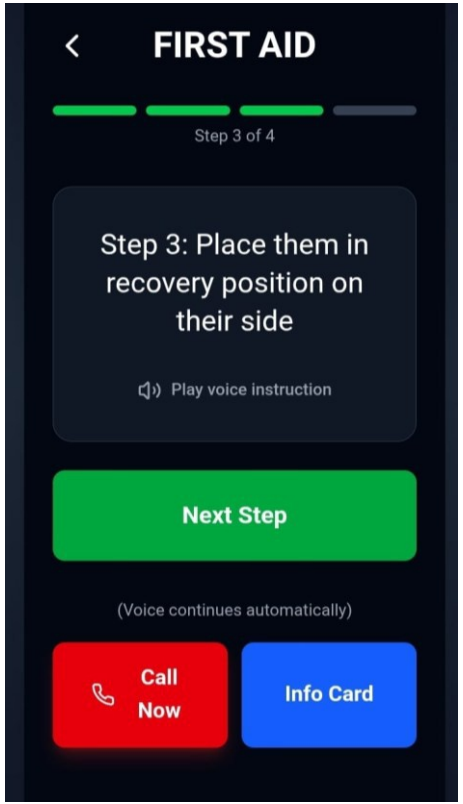

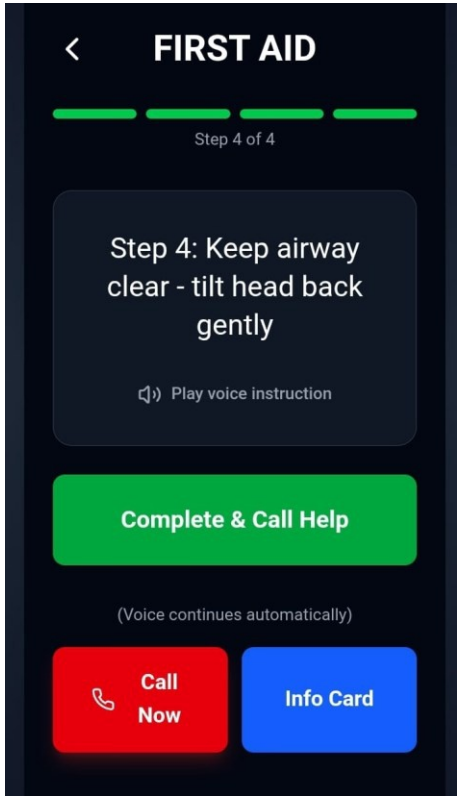

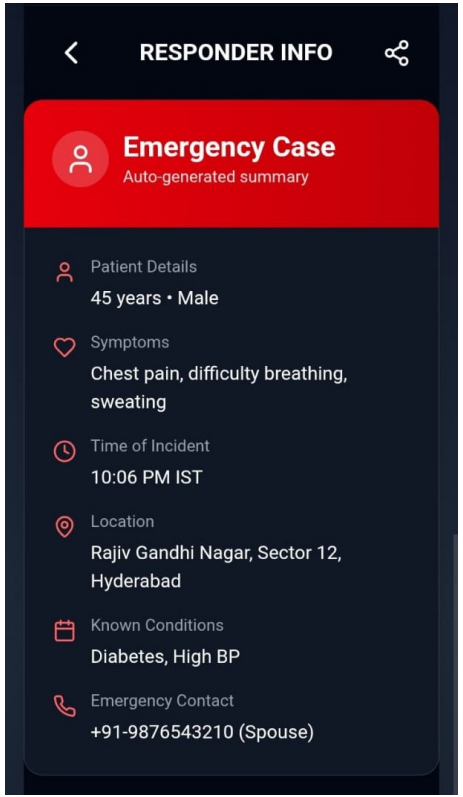

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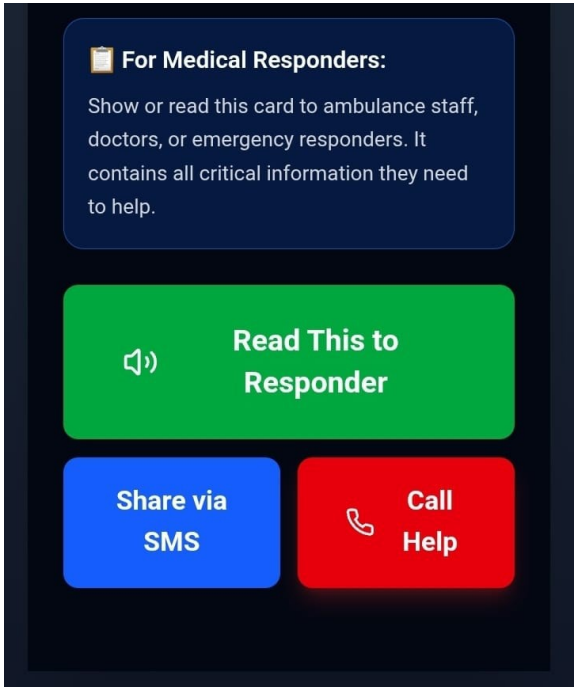
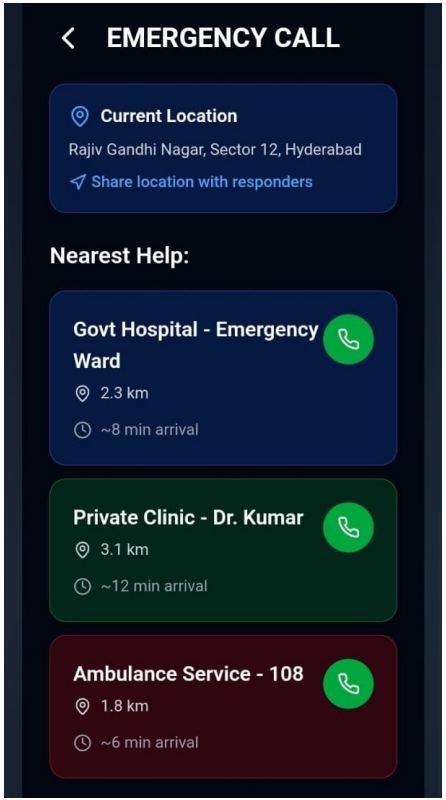
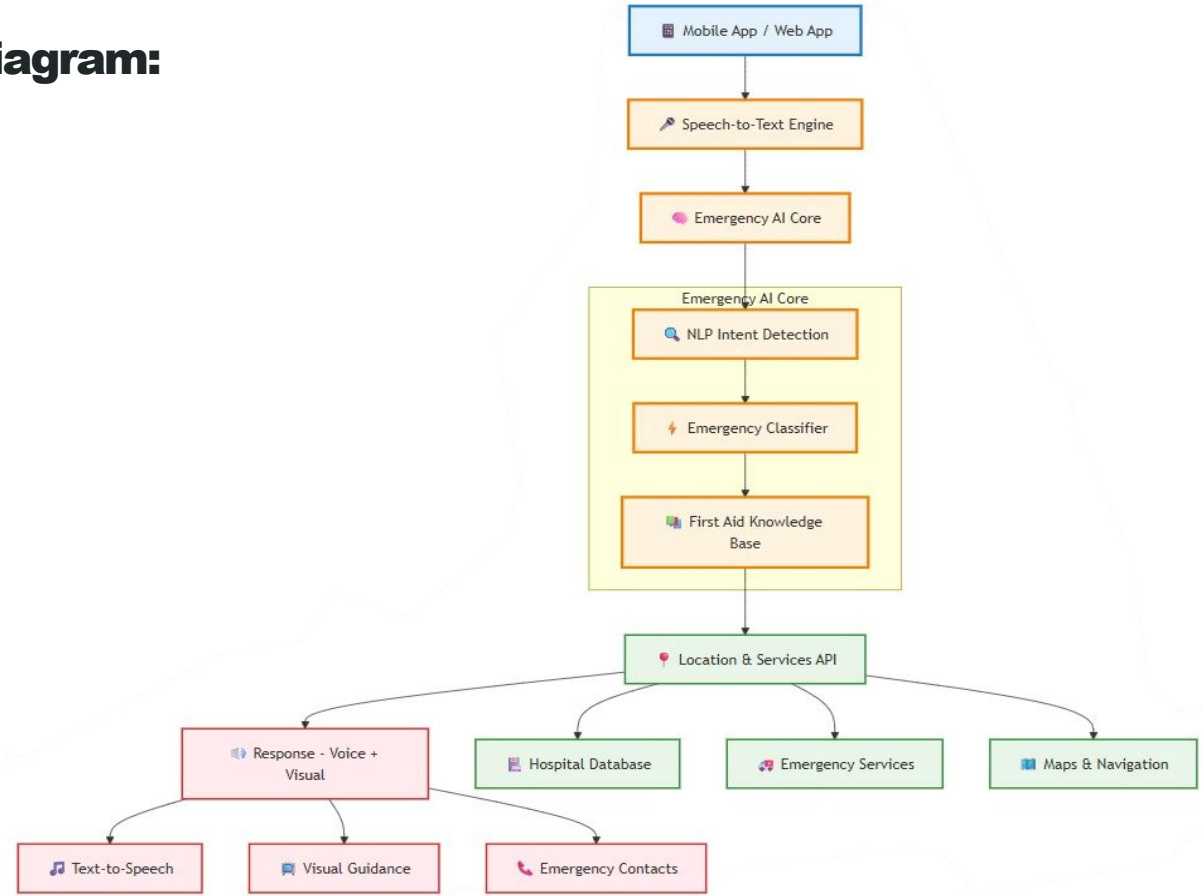
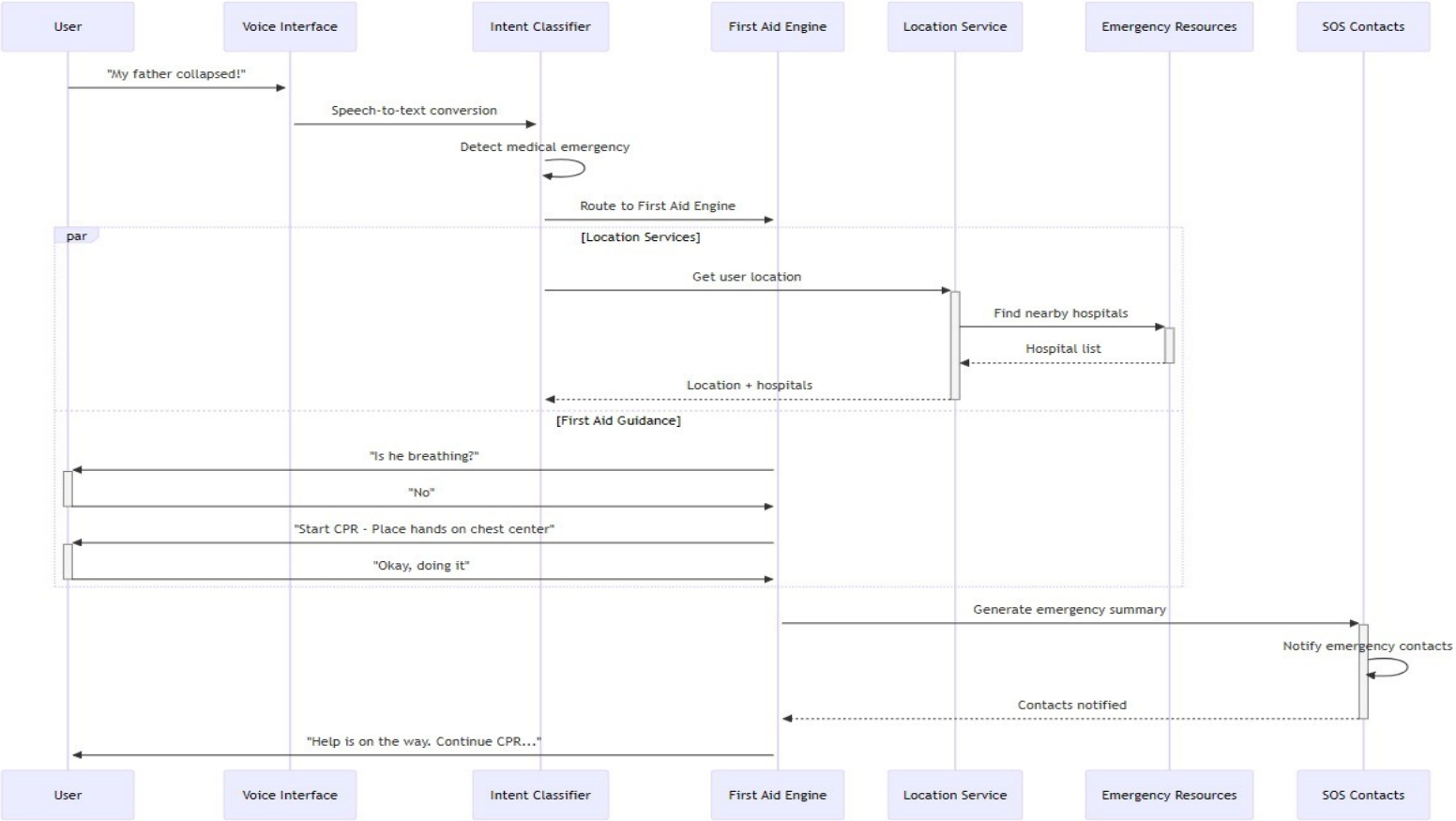




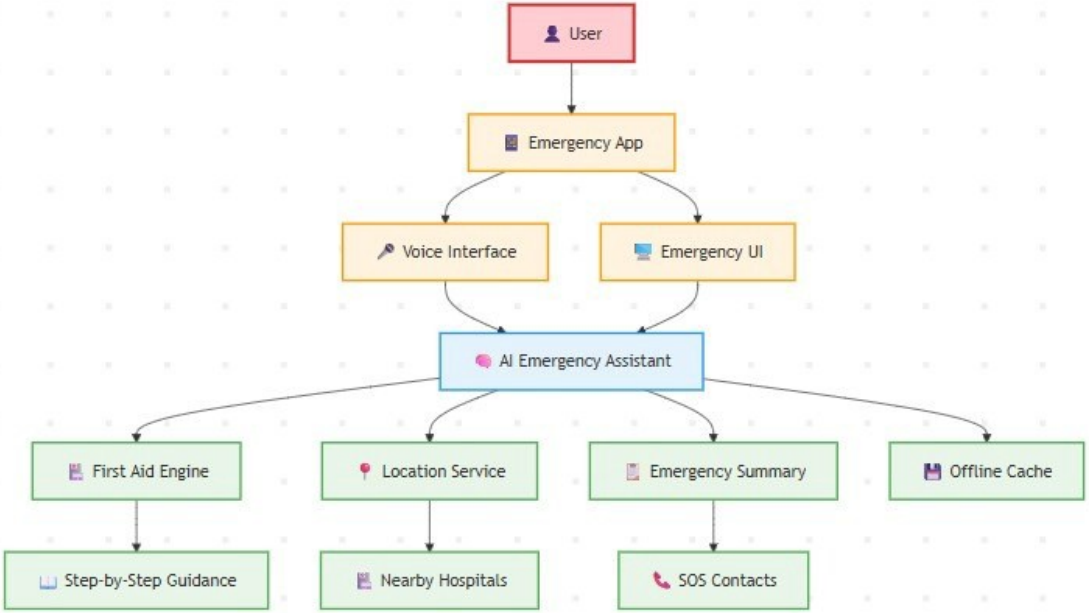
Architecture diagram:



Sequence diagram:



FlowChart diagram:



Target Users

- Rural & semi-urban populations
- Elderly caregivers
- Factory & construction workers
- Women & children in safety threats
- Disaster-prone communities

Example User Flow

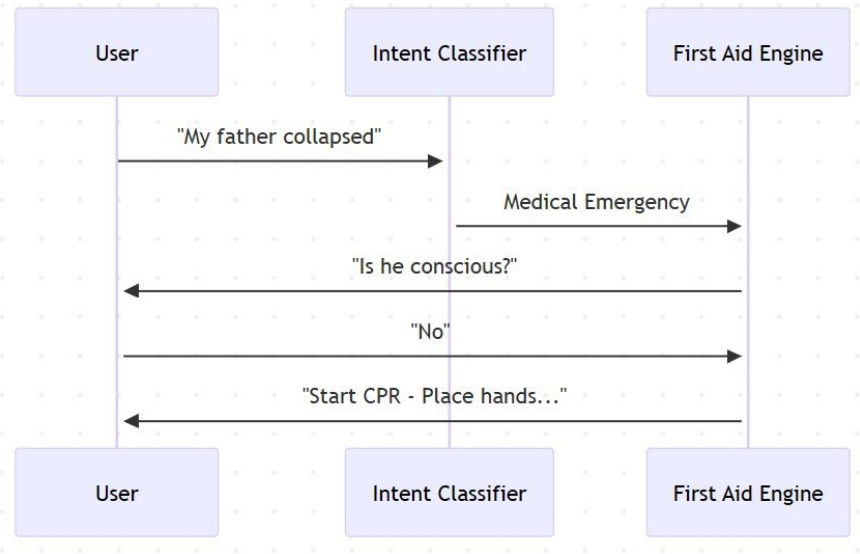
User: "My father collapsed"

AI Flow:

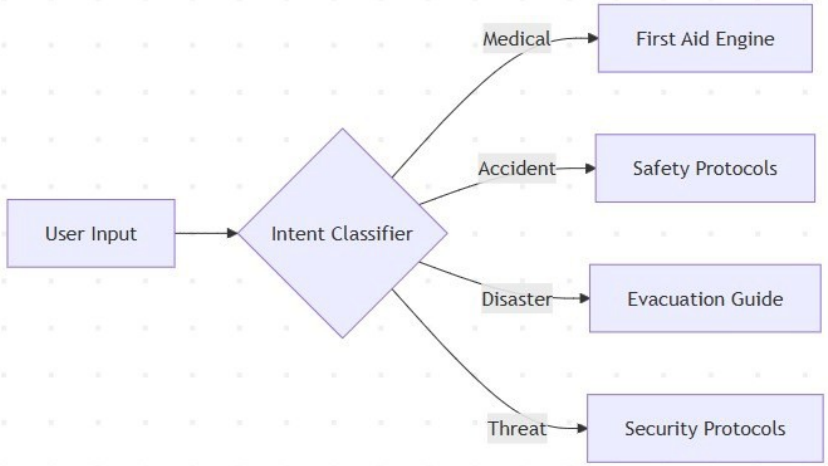
1. Detects medical emergency
2. Asks: "Is he conscious?"
3. Guides CPR or recovery position
4. Shows ambulance & hospital
5. Generates info card:

- Age
- Symptoms
- Time of incident

Sequence diagram:



Activity diagram:



Technologies to be used in the solution:

Frontend

1. Flutter / React Native
2. Android SDK
3. Accessibility UI Components

Backend

1. Python (Fast API)
2. Emergency Rules Engine
3. Vector DB for protocols

AI & ML

1. NLP intent classification
2. Speech-to-text(Google / Whisper)
3. Text-to-speech(Festival / Azure / Google)
4. Rule-based + ML hybrid for first aid logic

APIs

1. Maps & location services
2. Emergency contact database

Real-World Impact:

i. Saves lives in golden hour
ii. Helps rural &underserved communities
iii. Reduces panic-related mistakes
iv. Supports emergency responders with clear summaries
v. Scalable nationwide

Additional Details / Future Development

Future improvements include:

- Integration with national emergency helplines
- Wearable device integration for health monitoring
- Real-time disaster alerts
- Advanced AI for medical condition detection
- Support for more regional languages

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Thank You